1. One-Line Project Summary (30-second introduction)

Our project is an AI-powered platform that helps both recruiters and job seekers. Recruiters can predict a fair salary for a job posting, while candidates can upload their resume to check whether they are eligible for a job, identify missing skills, estimate the expected salary, and interact with a RAG chatbot for job-related questions.

2. What problem are we solving?

There are **two users**.

1. **Employer**

Problem:

- Doesn't know how much salary should be offered.
- May overpay.
- May underpay and lose candidates.

Solution:

➞ Predict market salary using Machine Learning.

2. **Candidate**

Problem:

- Doesn't know whether they fit a job.
- Doesn't know expected salary.
- Doesn't know missing skills.

Solution:

Upload resume →

AI analyzes it →

Compares with job →

Shows

- Match percentage
- Missing skills
- Experience fit
- Expected salary

3. Dataset

Guide may ask:

Which dataset?

Answer:

We are using the **LinkedIn Job Postings 2023–2024 Dataset**.

It originally consists of

- 11 CSV files

These files contain

- Job postings
- Skills
- Salary
- Companies
- Benefits
- Industries
- Employee count
- Company specialities

We merged them into

One master dataset

- **123,849 rows**
- **37 columns**

➢ **Overall project overflow –**

LinkedIn Dataset

↓

Merge 11 CSV files

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Data Cleaning

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Feature Engineering

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Machine Learning Models

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Salary Prediction

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Resume Parsing

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Skill Extraction

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Resume vs Job Matching

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Expected Salary

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Missing Skill Analysis

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RAG Chatbot

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Web Application

5. What ML tasks are we implementing?

There are **4 ML tasks**.

Task 1

Salary Prediction

Type:

Regression

Input

- Skills
- Experience
- Company

- Location
- Job Description

Output

Predicted Salary

Task 2

Resume Eligibility Prediction

Type

Multi-class Classification

Predicts

Experience Level

Example

Entry

Associate

Senior

Director

Executive

Intern

This is selected as the primary classification target because it is more balanced and useful than other possible targets.

Task 3

Resume Matching

NLP

Extract skills

Compare

Resume

VS

Job Description

Output

Match %

Missing Skills

Task 4

Job Clustering

Unsupervised Learning

Groups similar jobs together

6. Where is AI used?

Guide may ask

Why is this AI-powered?

Answer

AI is used in three places.

1.

Resume Parsing

Extract

- Skills
- Experience
- Education

using NLP.

2.

Generative AI

Generate explanation.

Example

Instead of

Salary = 14 LPA

AI says

This salary is recommended because the role requires Python, Spark, AWS, 5 years experience and is located in Bangalore.

3.

RAG Chatbot

Instead of ChatGPT answering from memory,

It searches real job postings.

Then answers.

So answers are

- More accurate
- Explainable
- Based on real data

The chatbot uses **FAISS** as a vector database to retrieve similar job postings before generating an answer.

7. What is RAG?

Simple answer

Normally

LLM

↓

answers from memory.

RAG

↓

Searches database

↓

Finds similar jobs

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Provides them to LLM

↓

LLM answers.

Hence

Hallucination reduces.

Accuracy increases.

8. What is FAISS?

Facebook AI Similarity Search

Stores

Embedding vectors.

When user asks

Salary for Python Developer in Pune

FAISS finds similar jobs.

Then LLM generates answer.

- **Which preprocessing?**

- Remove duplicates
- Handle missing values
- Text cleaning
- Merge skills
- Feature engineering
- Encoding categorical variables
- Train/Test split
- Class balancing using SMOTE

The project documentation recommends class_weight='balanced' and testing SMOTE for minority classes.

11. Which algorithms?

Regression

Predict Salary

Possible models

- Linear Regression (Baseline)
- Random Forest Regressor
- XGBoost Regressor
- CatBoost Regressor

Classification

Predict Experience Level

Models

- Logistic Regression
- Random Forest
- XGBoost
- BERT (Deep Learning)

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12. Evaluation Metrics

Regression

- MAE
- RMSE
- R^2 Score

Classification

- Accuracy
- Precision
- Recall
- Macro F1-score
- Confusion Matrix

Macro F1 is preferred because the classes are imbalanced.

16. Expected Outputs

For Candidate

- Resume score
- Skill match %
- Missing skills
- Experience level prediction
- Salary estimate
- AI explanation
- Chatbot

For Recruiter

- Suggested salary
- Salary justification
- Similar jobs
- Chatbot assistance

Question	Short Answer
What is the main objective?	Predict fair salaries and analyze resumes for better hiring decisions.
Why LinkedIn dataset?	It contains real-world job postings, salaries, skills, companies, and locations.
Why merge 11 files?	To combine all job, company, skill, salary, and industry information into one dataset.
Why Regression?	Salary is a continuous numeric value.
Why Classification?	Experience level is a categorical target with six classes.
Why NLP?	To understand job descriptions and resumes.
Why RAG?	To answer questions using real job postings instead of only the LLM's internal knowledge.
Why FAISS?	To efficiently retrieve similar job postings using vector similarity.
Why SMOTE?	To balance minority experience-level classes during training.
Final output?	Salary prediction, resume analysis, skill-gap detection, experience prediction, and an AI chatbot.

One-minute explanation you can confidently say

"Our project is an AI-Powered Salary Intelligence and Resume Analyzer Platform built using the LinkedIn Job Postings dataset. We first merge 11 CSV files into a master dataset with over 123,000 job postings. Then we preprocess the data and build machine learning models for two main tasks: salary prediction using regression and experience-level prediction using multi-class classification. On the candidate side, a resume is parsed using NLP, its skills are compared with a selected job description, and the system reports a match score, missing skills, expected experience level, and estimated salary. On the recruiter side, the system recommends a competitive salary for a new job posting along with an AI-generated explanation. Finally, a RAG chatbot backed by a FAISS vector database retrieves similar real job postings to answer user questions accurately, reducing hallucinations and making the platform more reliable."